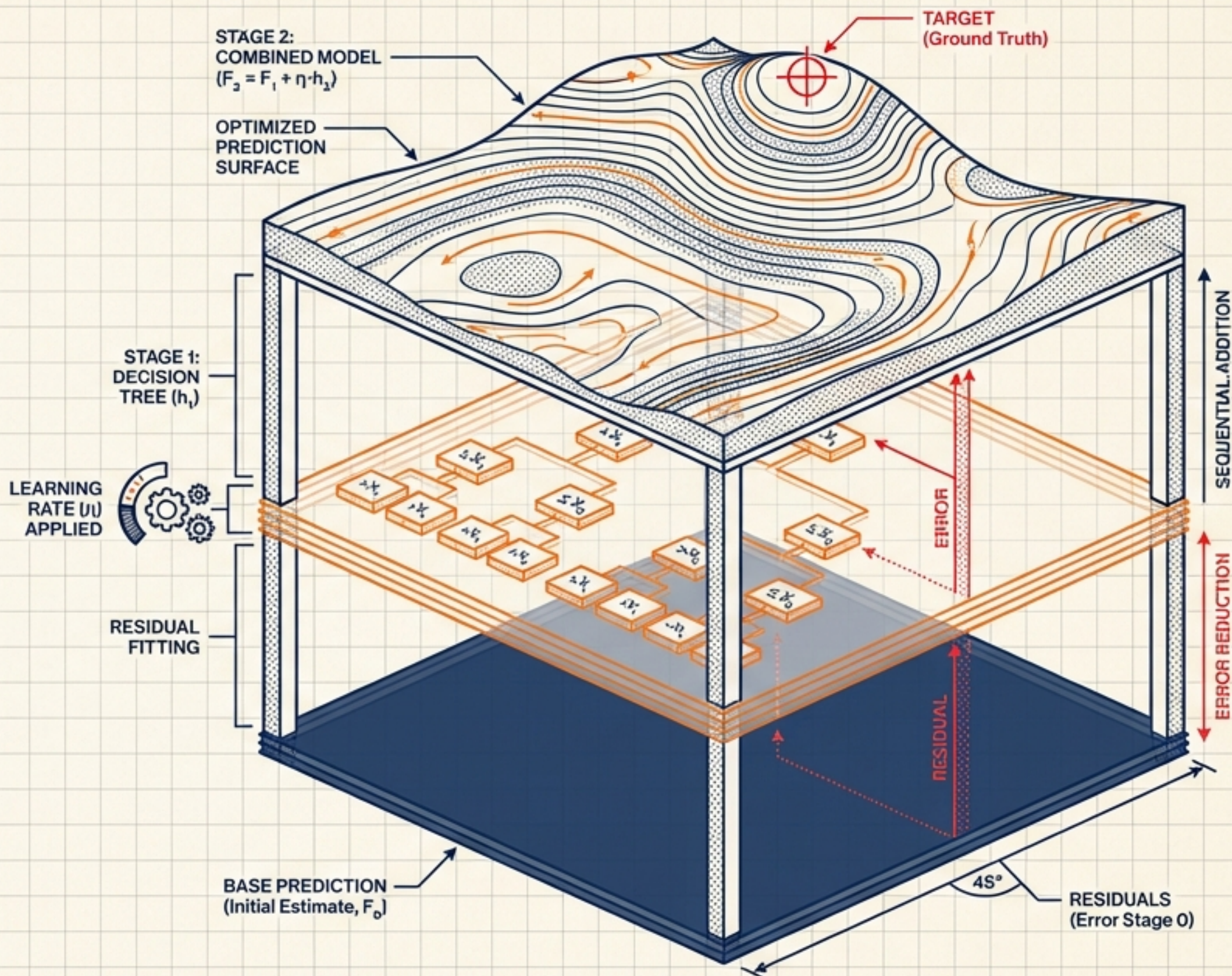


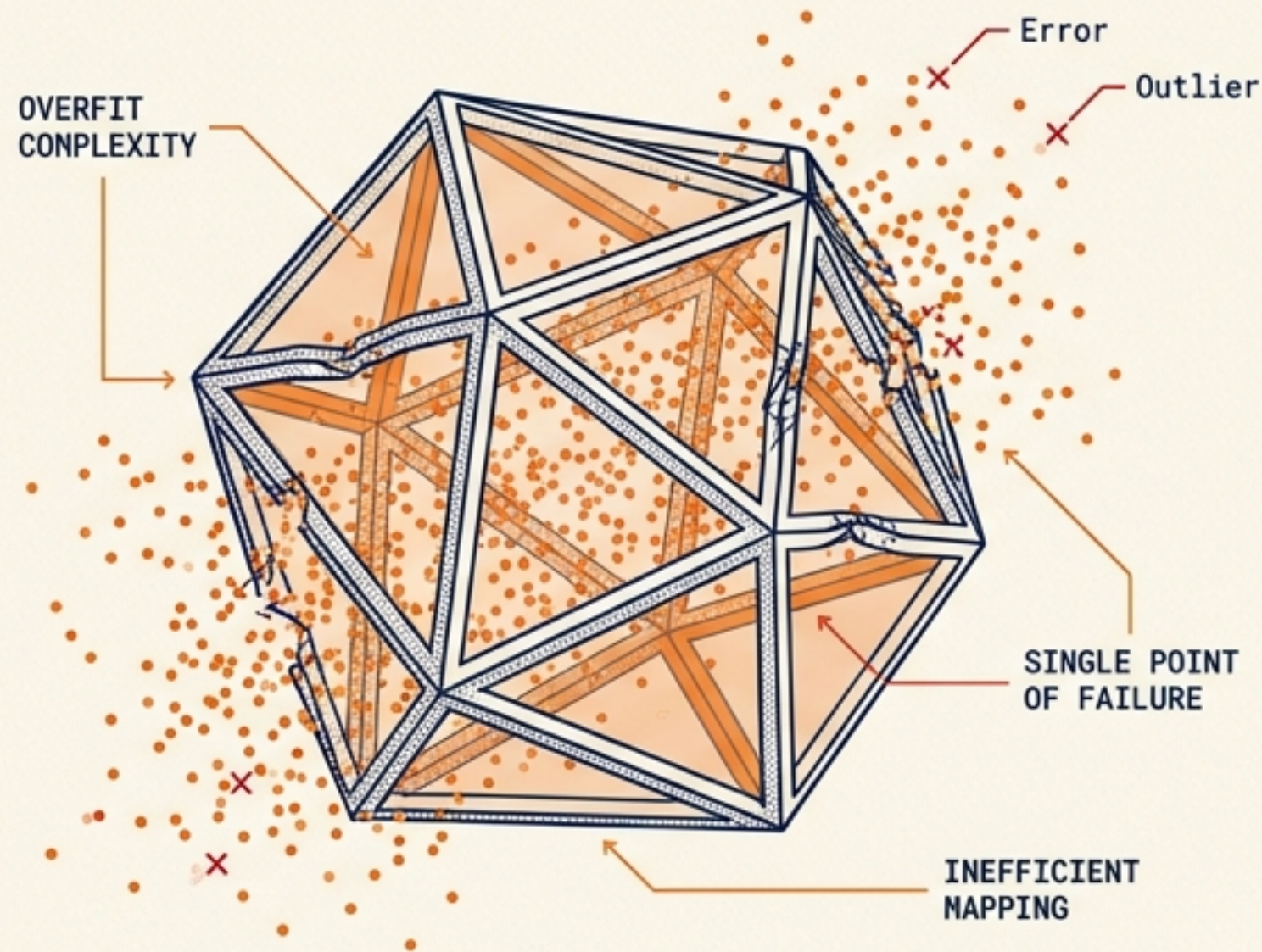
Decoding Gradient Boosting.

An architectural breakdown of sequential stage-wise machine learning.



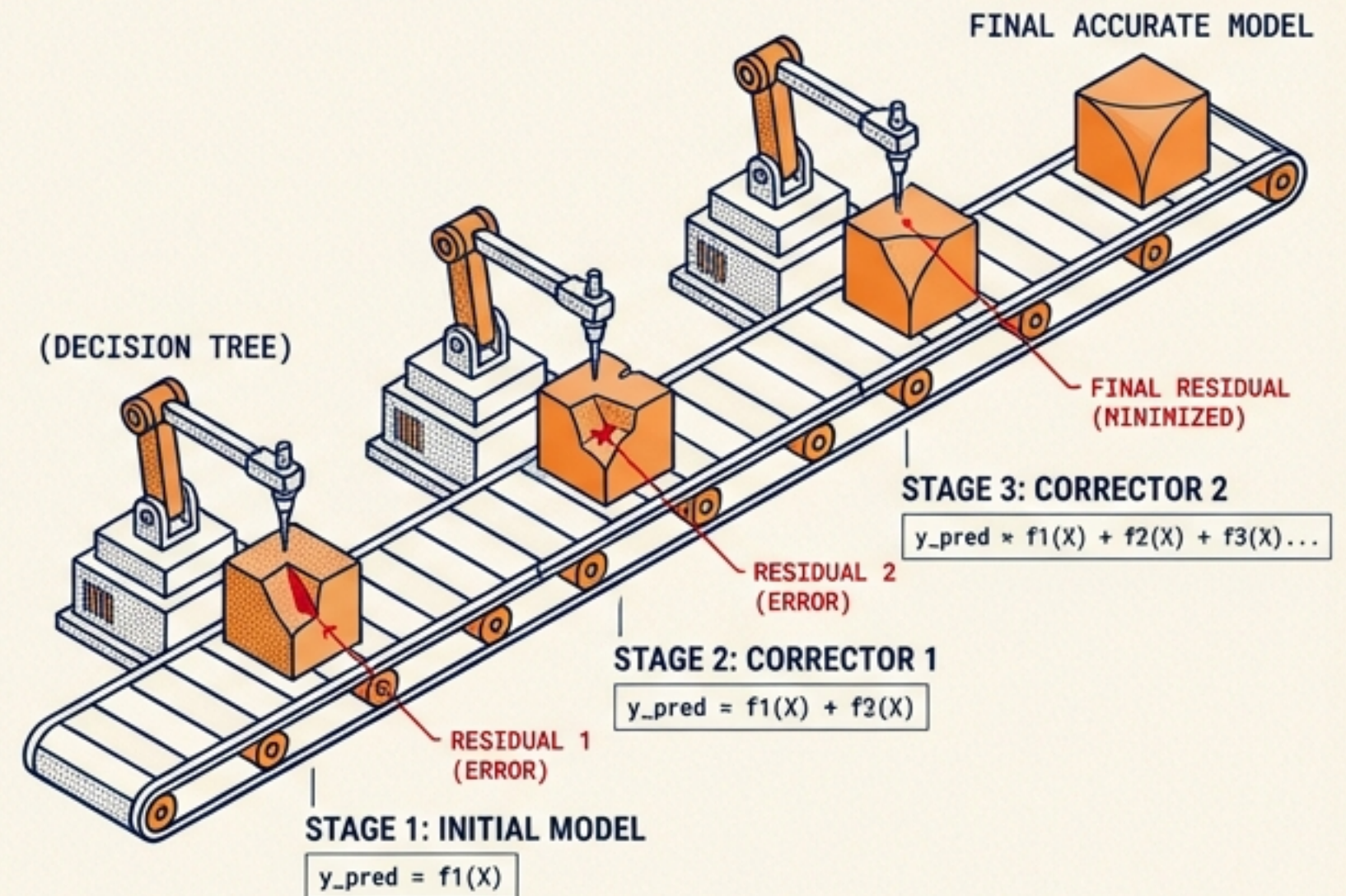
The Assembly Line of Correctors.

Monolithic Models



A single algorithm attempts to map the entire complexity of the data at once.

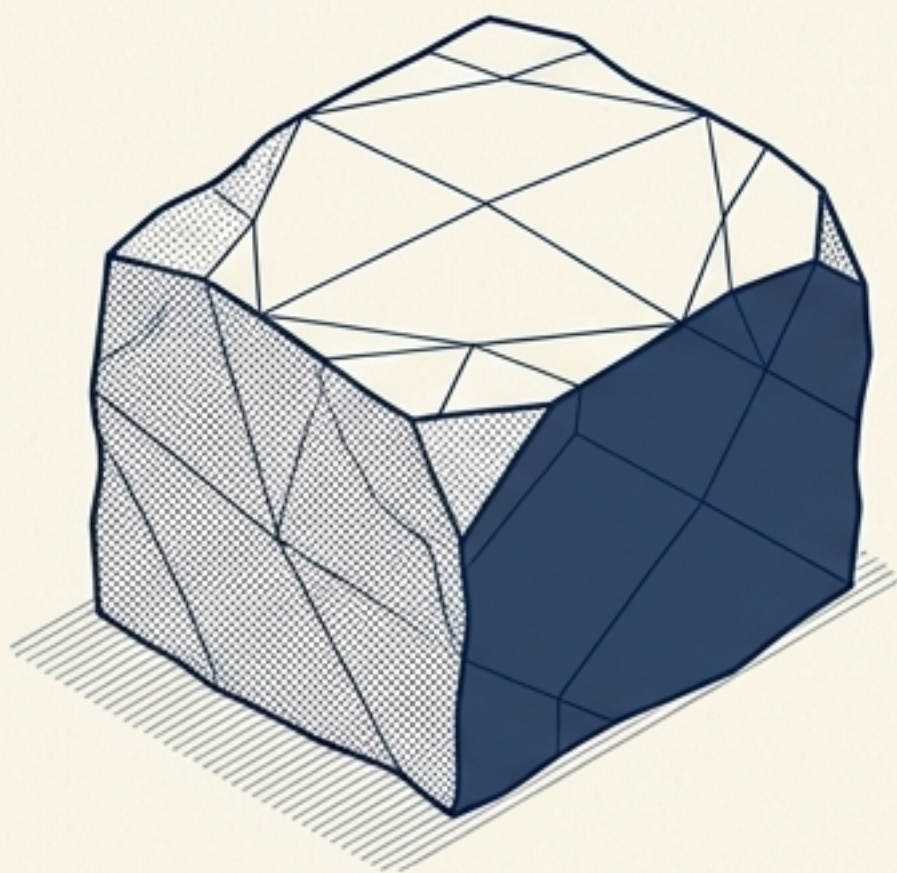
Sequential Stage-Wise Addition



A cascade of focused models, where each new addition exists purely to fix the mistakes of the previous one

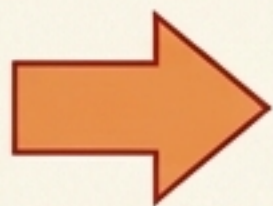
The Iterative Sculpting Metaphor.

Gradient boosting leaves the existing structure intact and adds a new layer designed exclusively to carve away remaining errors.



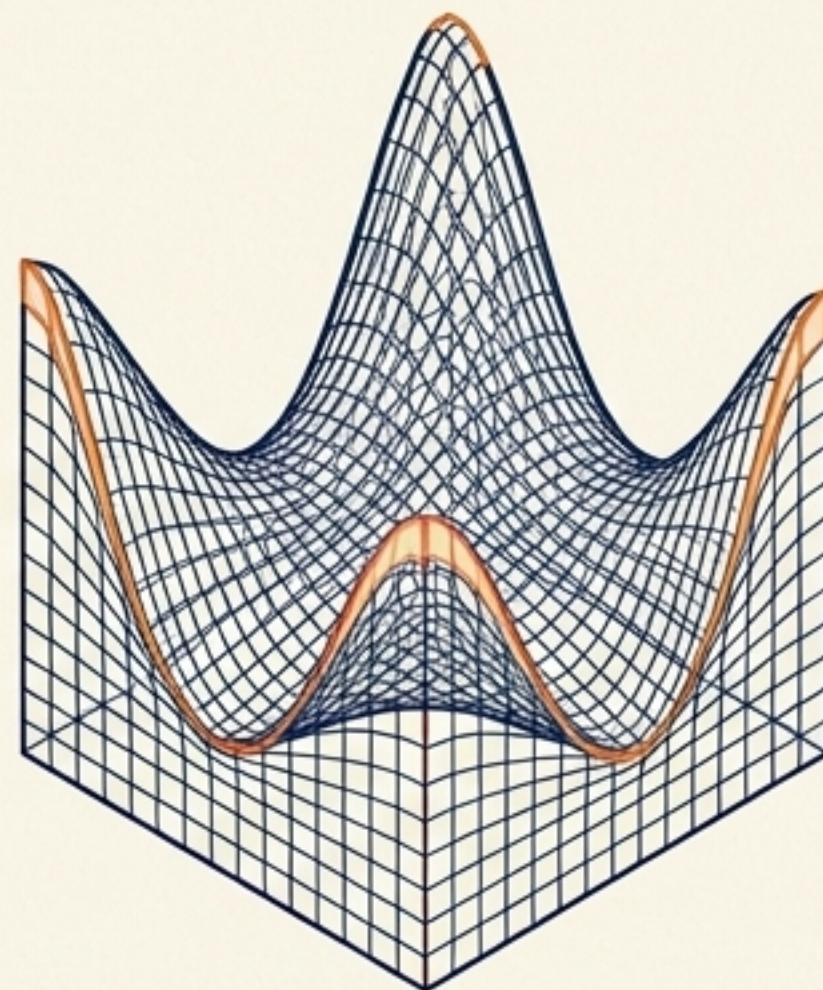
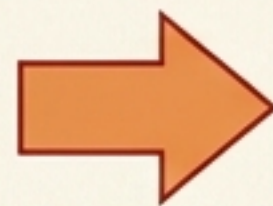
Stage 1: The Block.

Start with a blunt baseline
(The Mean).



Stage 2: The Chisel.

Target the exact imperfections
(Pseudo-residuals).



Stage 3: The Statue.

Layer multiple corrections
to achieve the final shape.

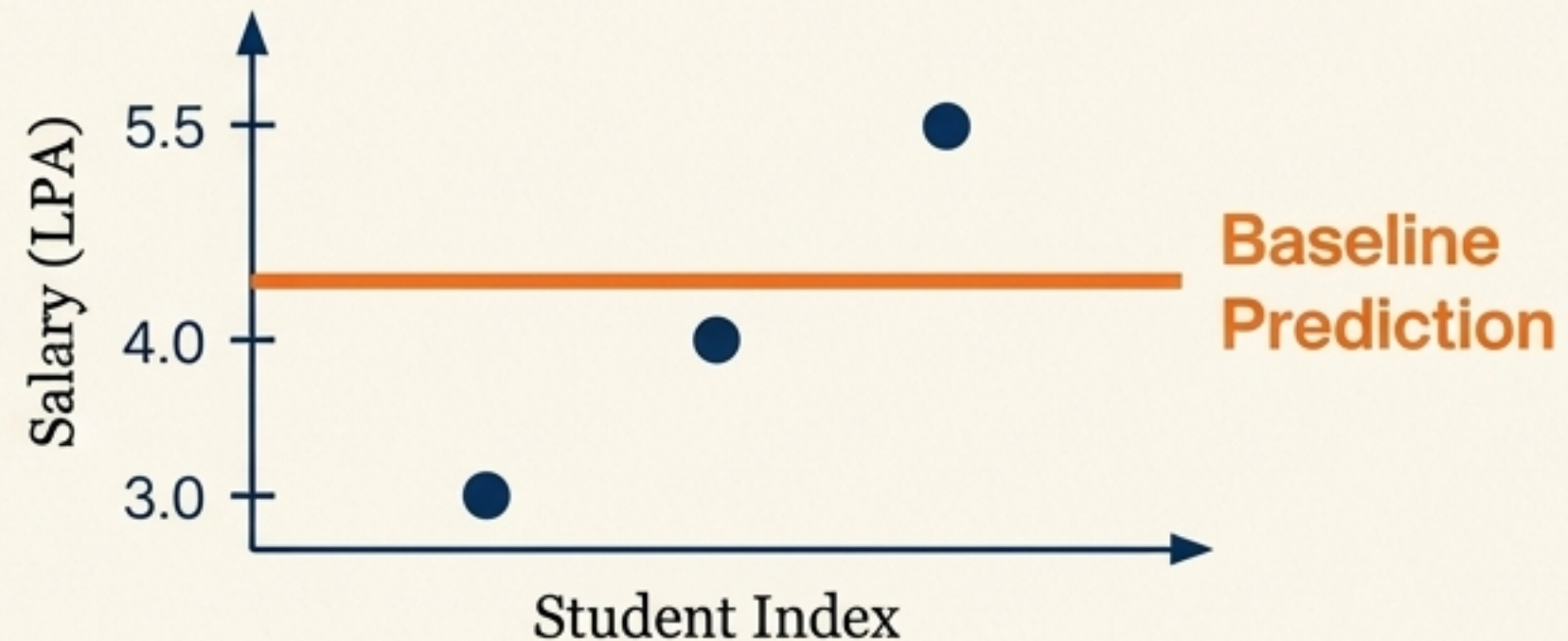
Step 0: The Dummy Baseline

In regression problems, Gradient Boosting starts with the simplest possible assumption: The Mean.

Model 1 ignores IQ and CGPA entirely. If asked for a prediction, it will blindly guess 4.3 LPA for every single student.

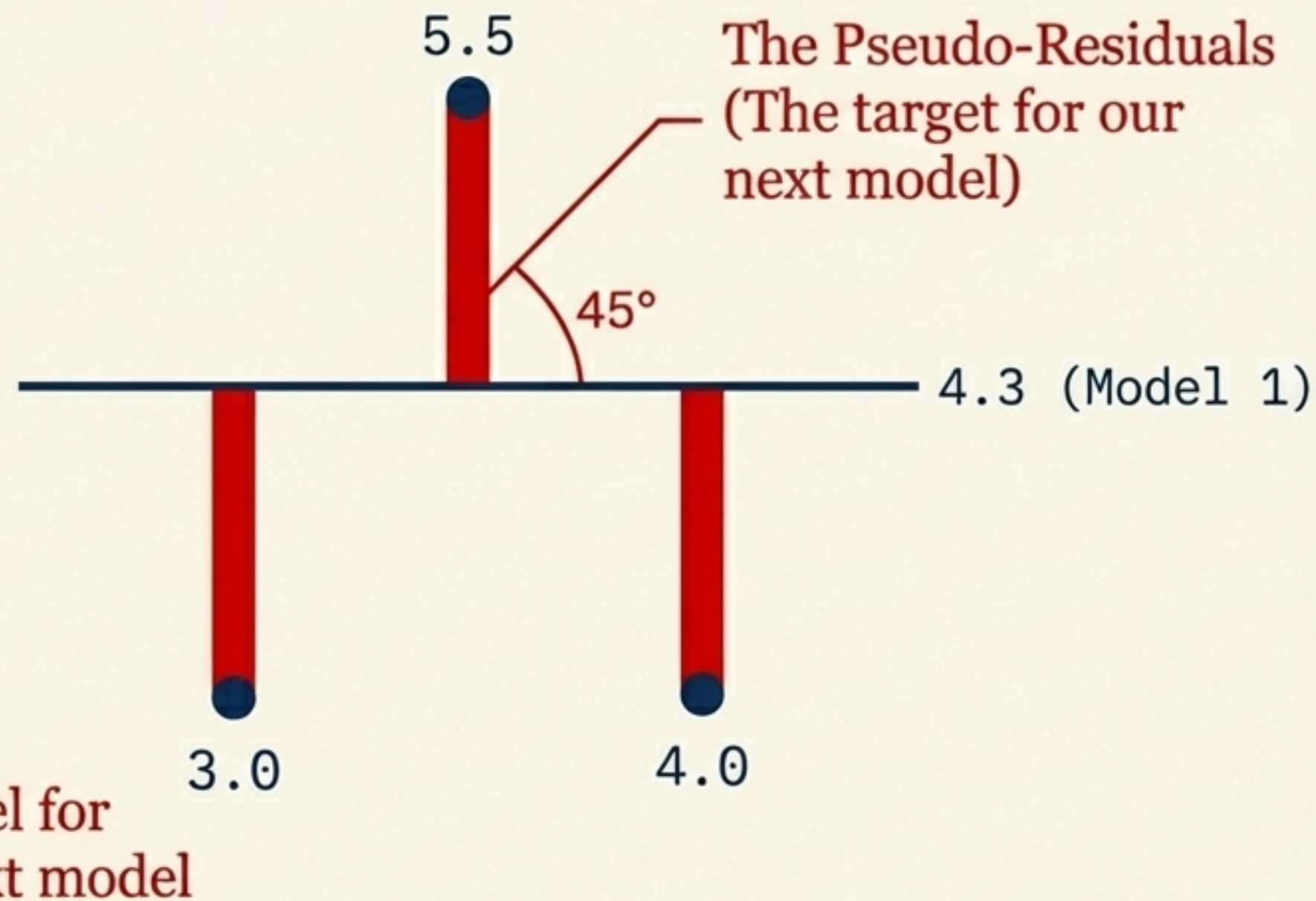
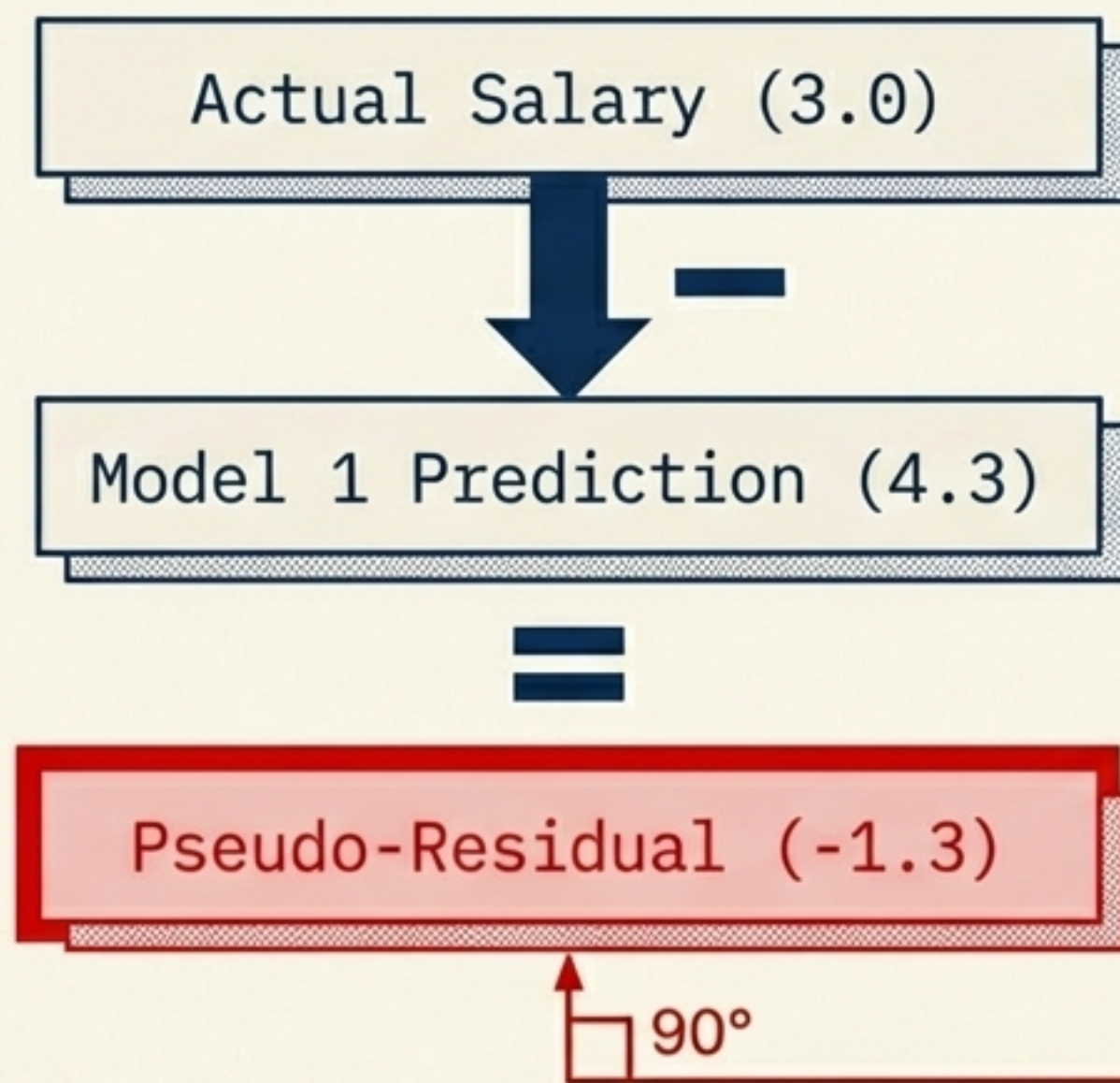
IQ	CGPA	Salary (LPA)
100	7.5	3.0
110	8.0	5.5
105	7.8	4.0

Mean = 4.3 LPA



Step 1: Isolating the Mistake.

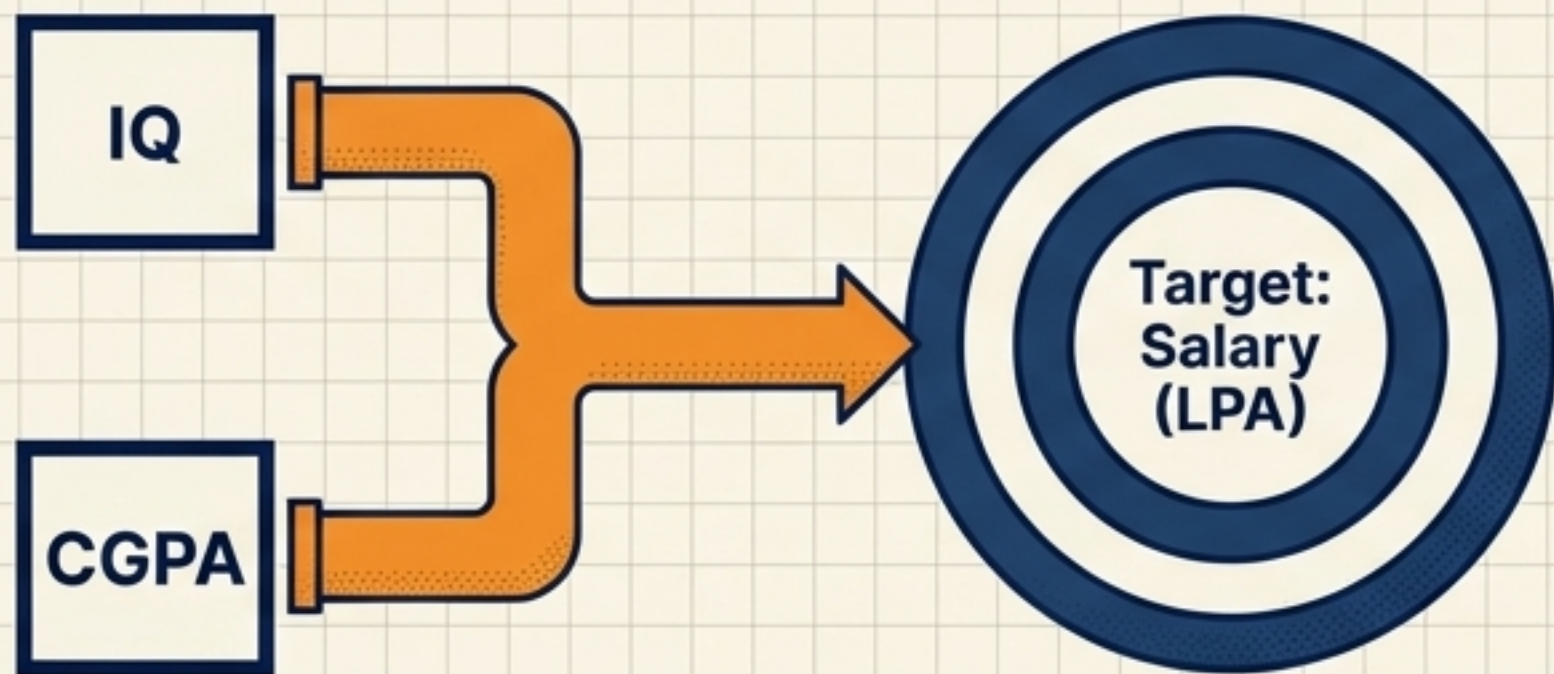
We calculate exactly how wrong our dummy baseline was.
This error becomes the fuel for the next stage.



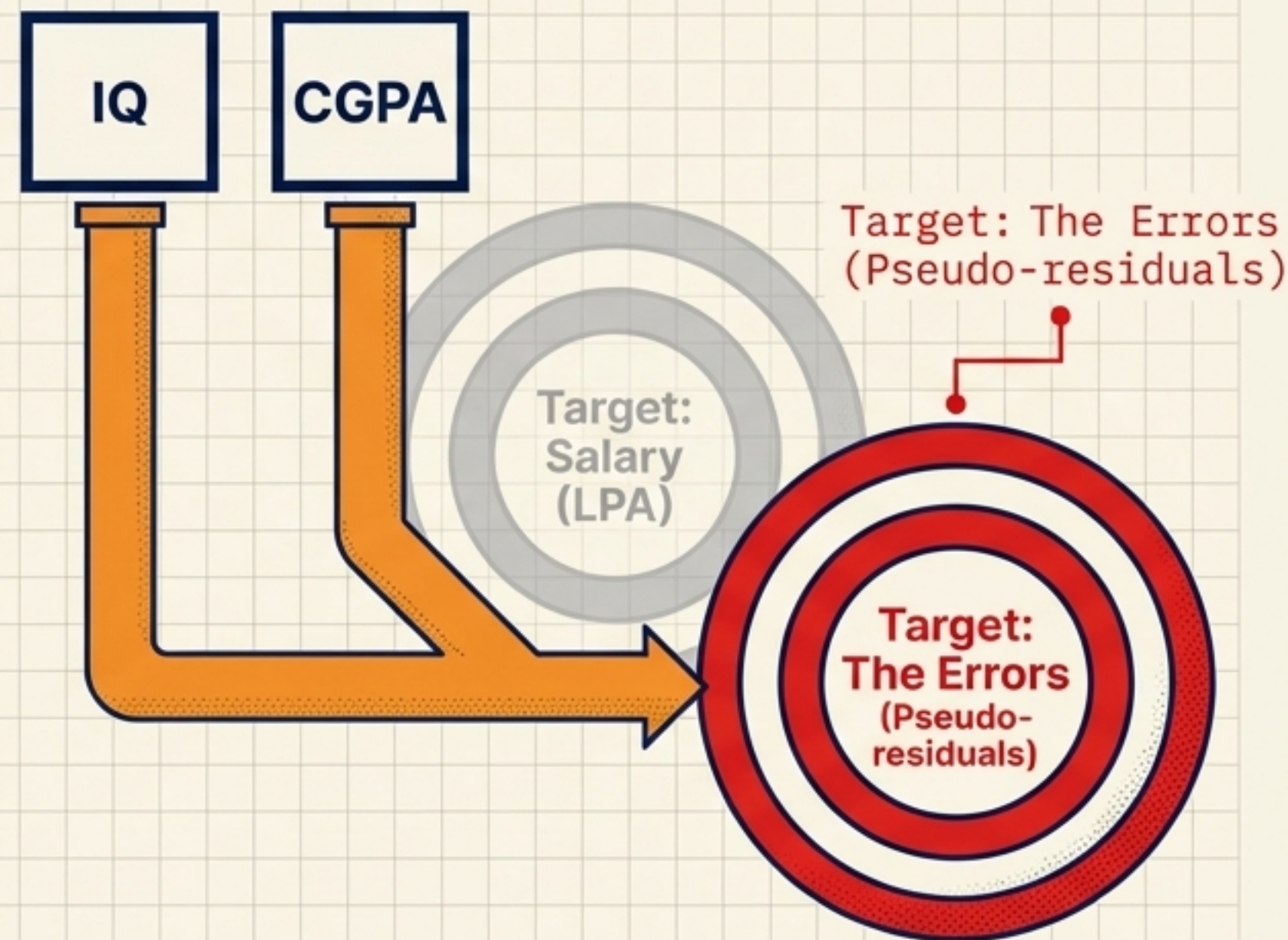
The Critical Pivot: Shifting the Target

Subsequent models do not predict the actual target. Model 2 is blind to Salary. It uses the inputs solely to predict how much Model 1 missed by.

Model 1's View

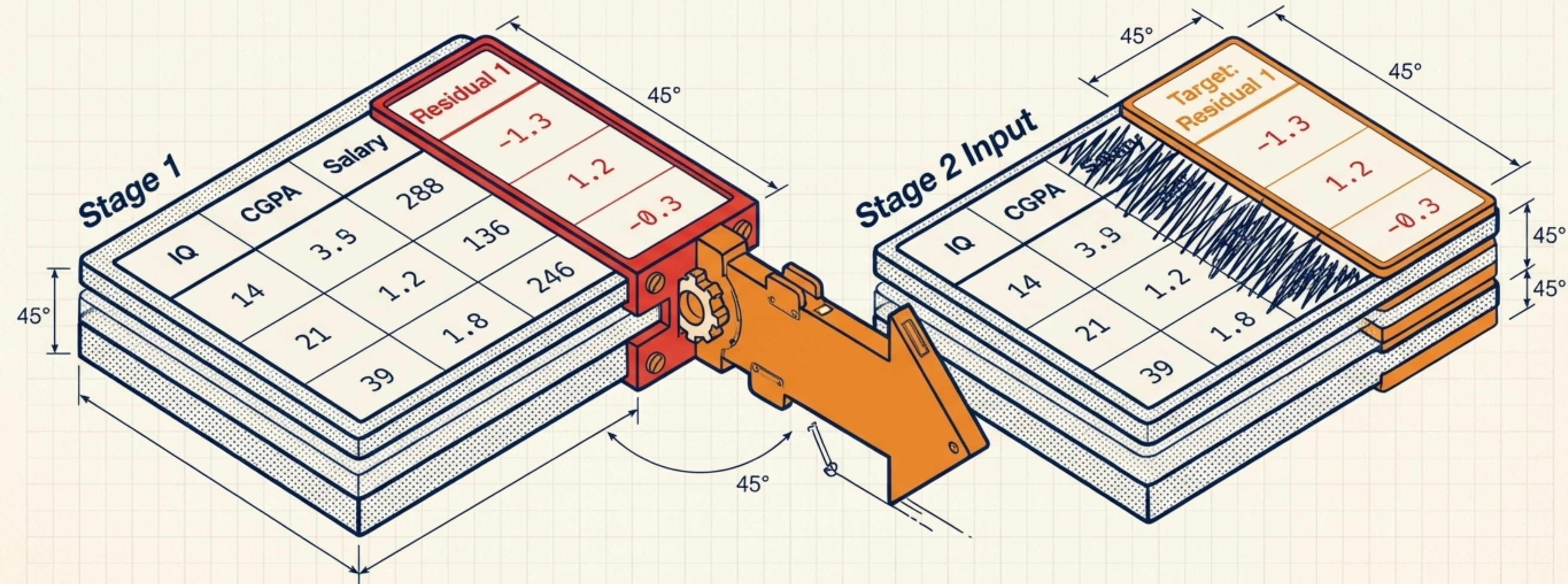


Model 2's View



The Cascading Data Flow

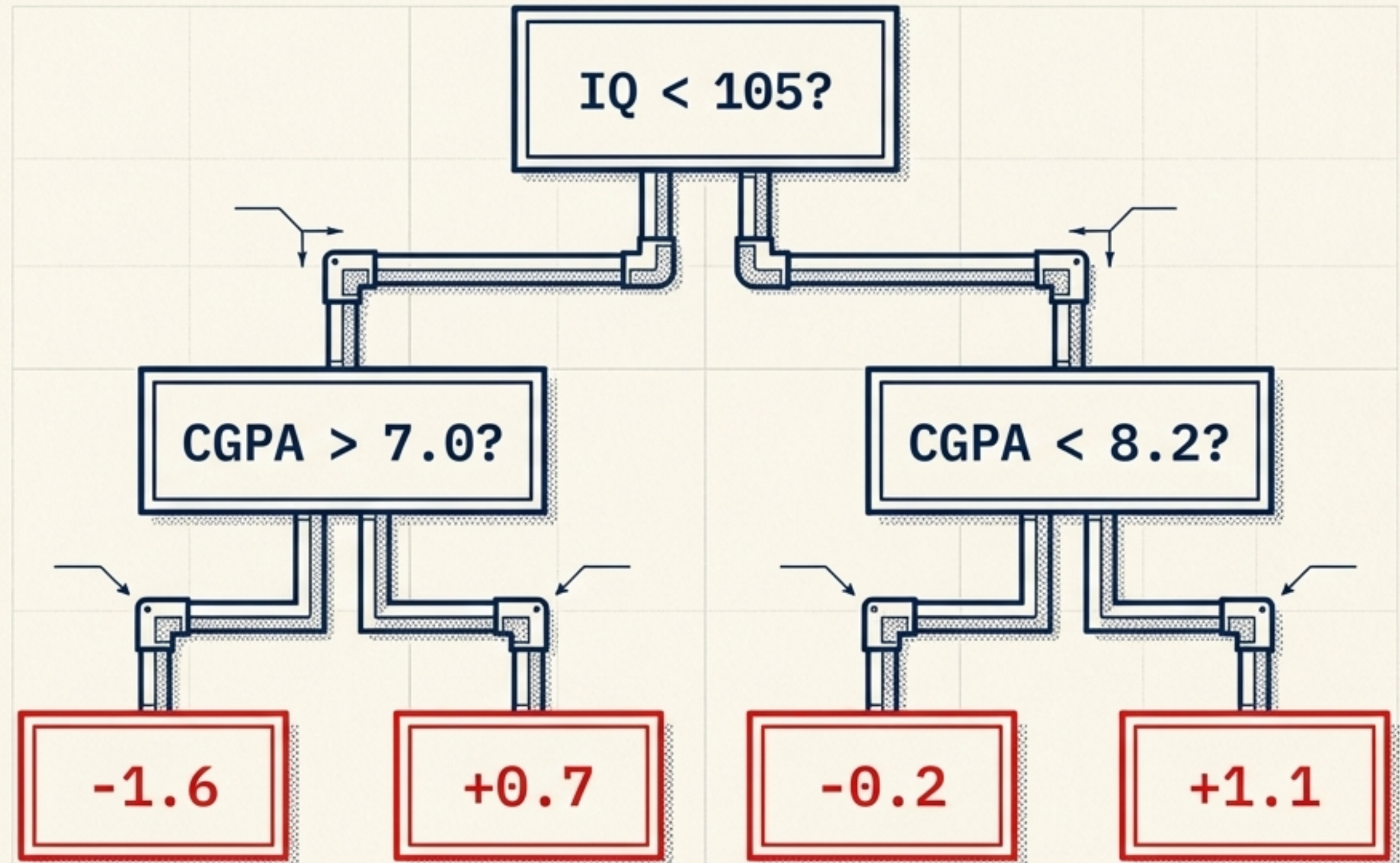
The output of Model 1's failure becomes the absolute truth (the target) for Model 2.



Training the Specialist (The Decision Tree).

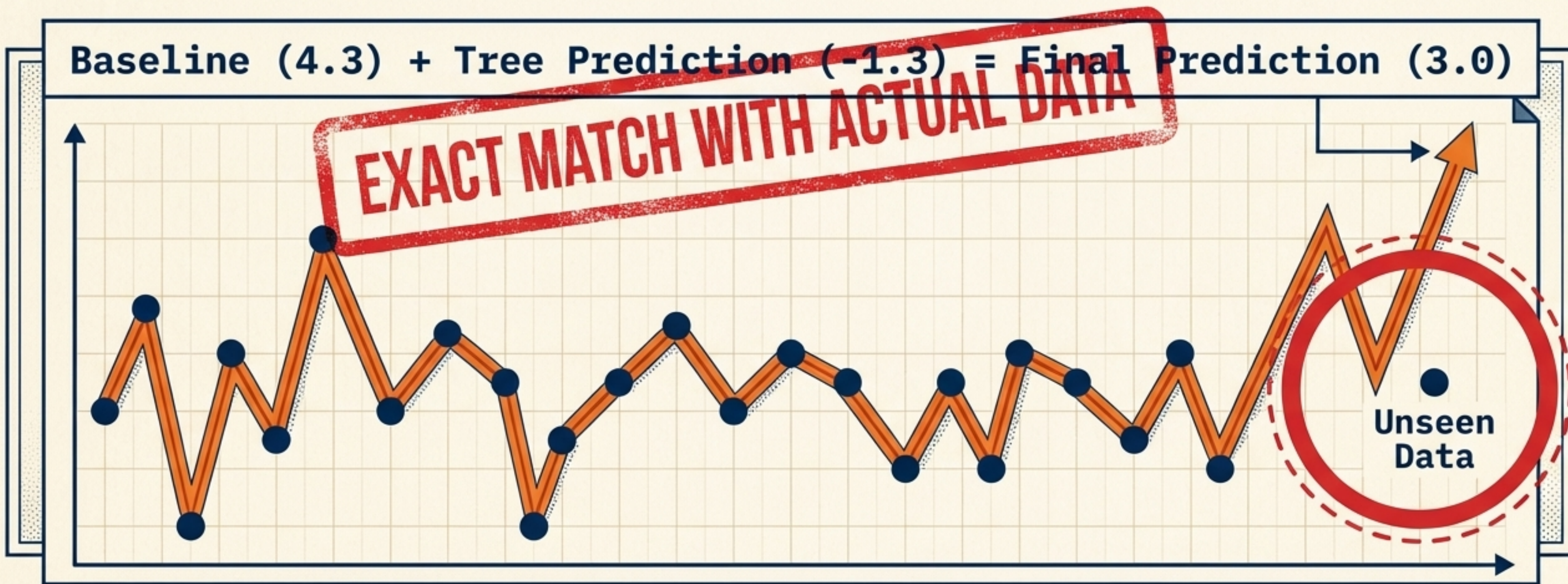
We train a Decision Tree on the new dataset to predict the residuals.

Unlike simple stumps, Gradient Boosting typically utilises deeper trees (8 to 32 max leaves) to capture complex, non-linear interactions in the errors.



The Danger of Perfect Memory

If we directly add Model 2's output to Model 1, we achieve perfect predictions on training data. This is Overfitting. It will fail spectacularly in the real world.



The Regulator: The Learning Rate Dial

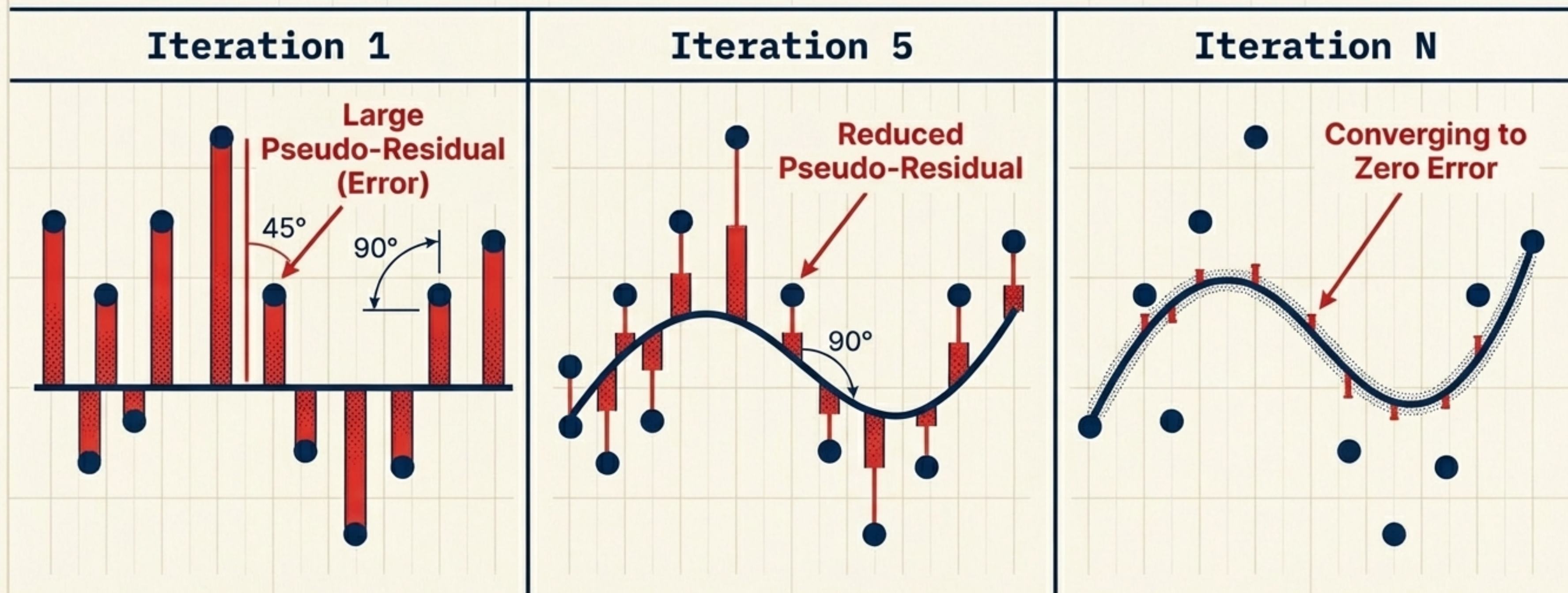
Instead of fully trusting the new model, we only accept a fraction of its advice. This forces the algorithm to take tiny, cautious steps towards the truth, ensuring robust generalisation.



$$\text{Prediction} = \text{Baseline} + (0.1 \times \text{Tree Prediction})$$

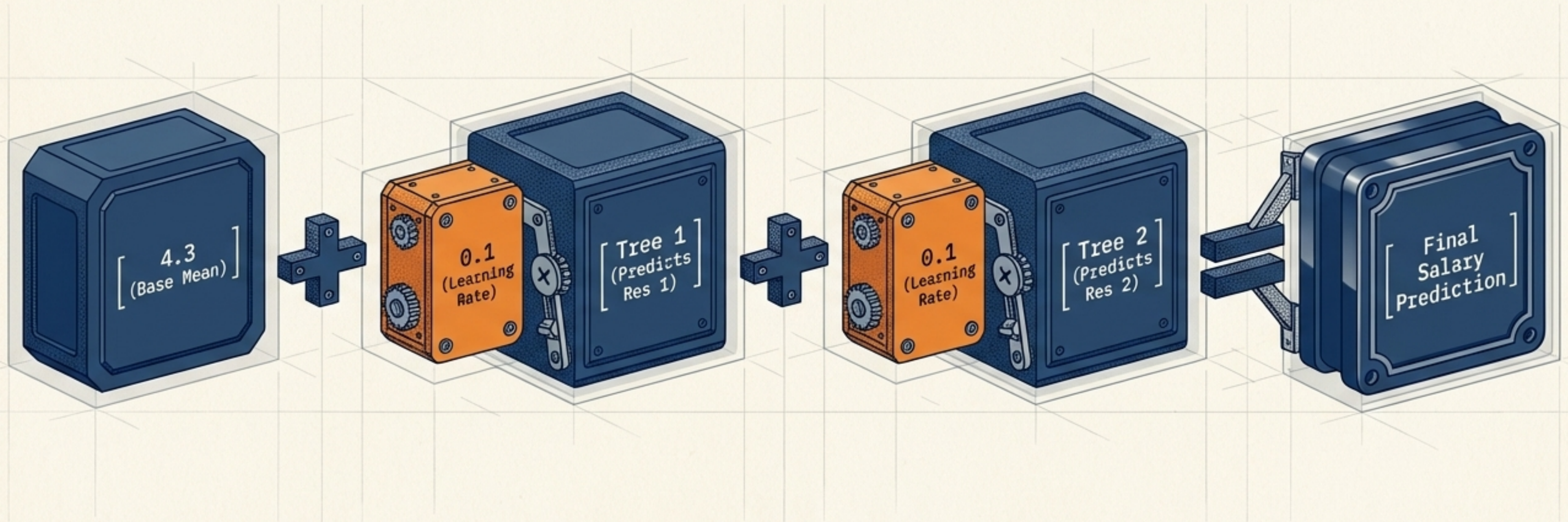
The Shrinking Error Bars

With every new model added at a fractional learning rate, the pseudo-residual mathematically converges towards zero. The mistakes get smaller and smaller.

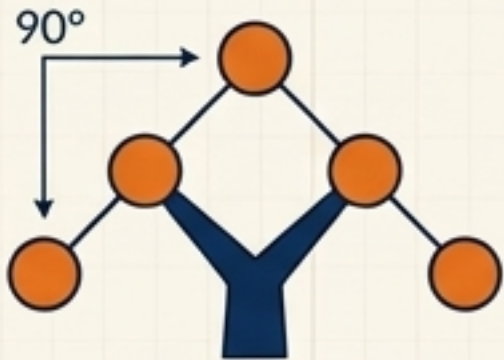
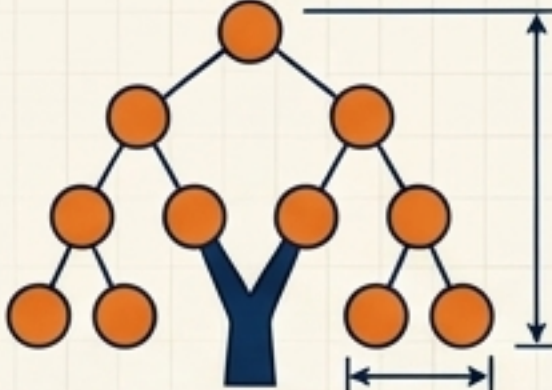
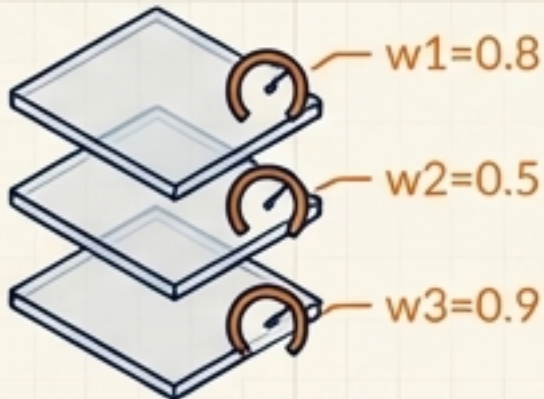
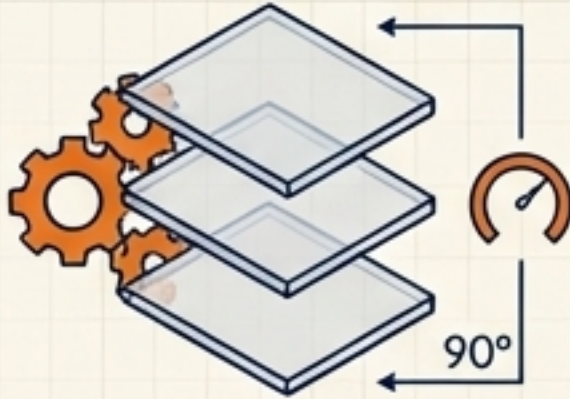
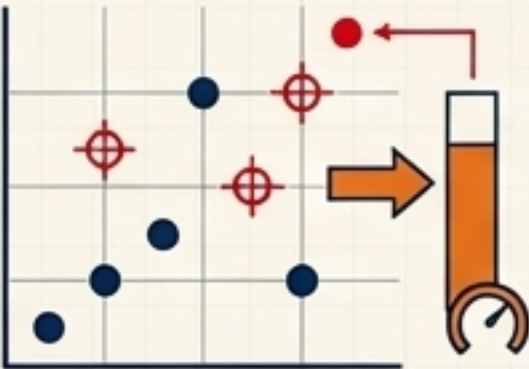
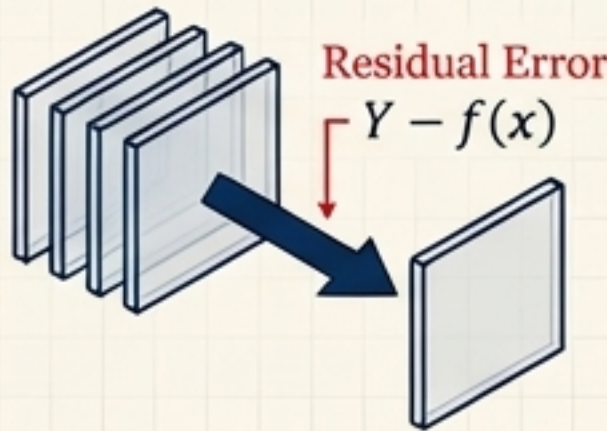


The Final Assembly

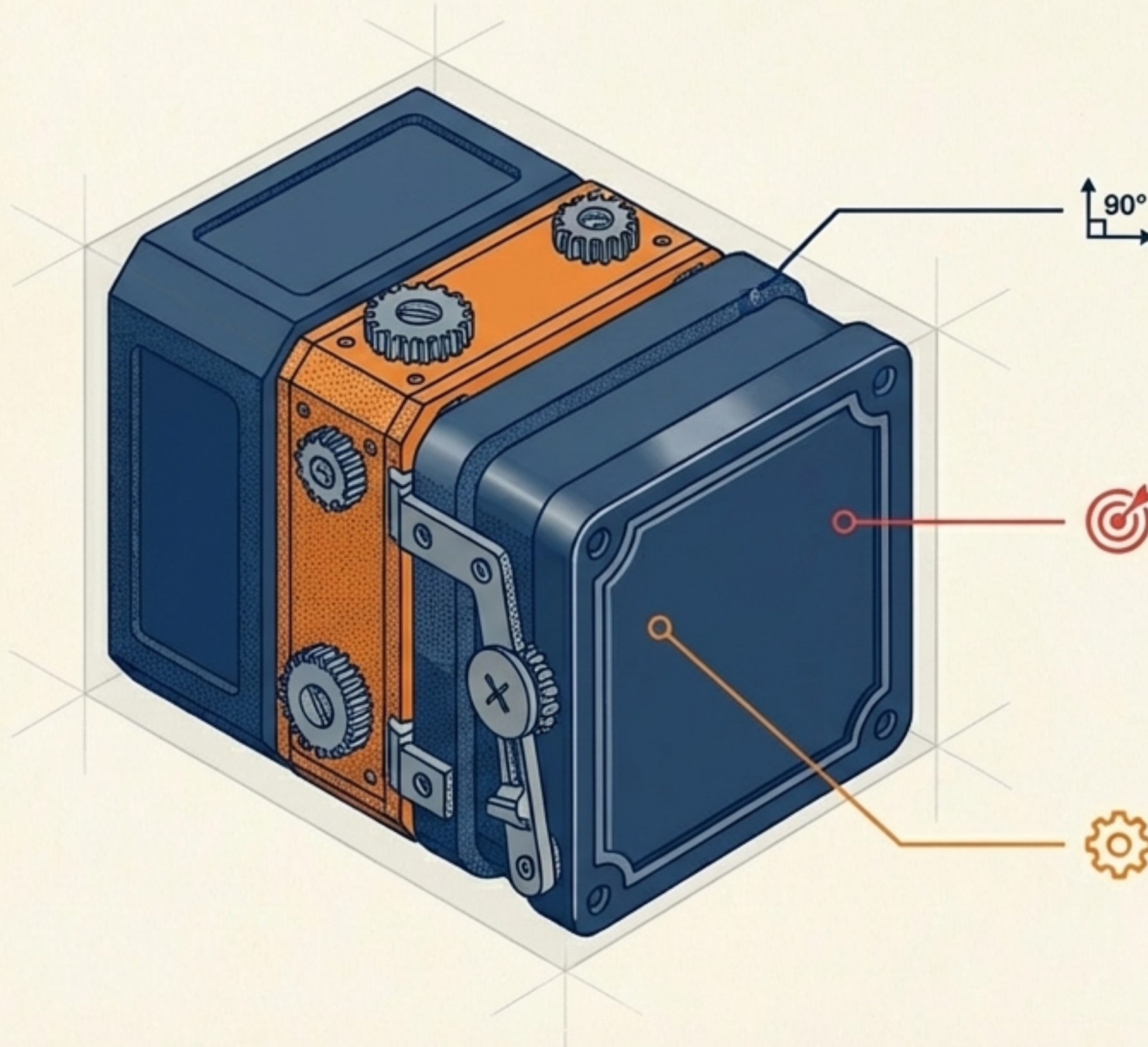
Gradient boosting is not a single complex equation.
It is a strictly controlled recipe of sequential corrections.



The Ensemble Showdown: AdaBoost vs. Gradient Boosting

	AdaBoost	Gradient Boosting
The Base Learner	Stumps (Max depth 1) 	Deeper Trees (Max leaves 8 to 32) 
The Weighting Scheme	Individual weights (w_1, w_2, w_3) for each specific model 	Constant universal Learning Rate (e.g., 0.1) for all models 
The Correction Mechanism	Increases the statistical weights of misclassified data rows 	Directly trains models to predict the pseudo-residual errors 

Master of the Blueprint.



Sequential, Not Simultaneous.

Models are built stage-wise; each relies entirely on the failures of the last.

Targeting Mistakes.

The algorithm uniquely trains models on the pseudo-residuals, never the original target.

Regulated by Design.

The learning rate acts as the ultimate braking mechanism, turning potential overfitting into robust generalisation.